



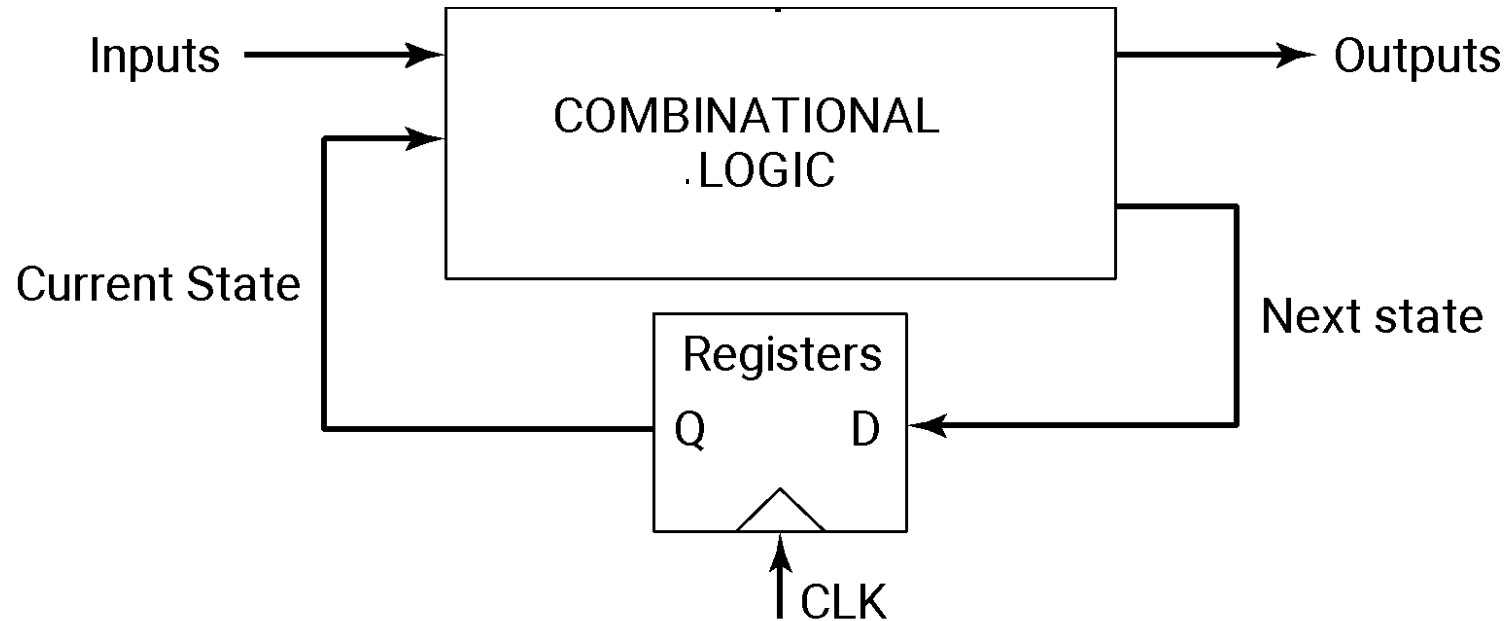
Digital Integrated Circuits

A Design Perspective

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Anantha Chandrakasan
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Timing Issues

Sequential Logic



2 storage mechanisms

- positive feedback
- charge-based

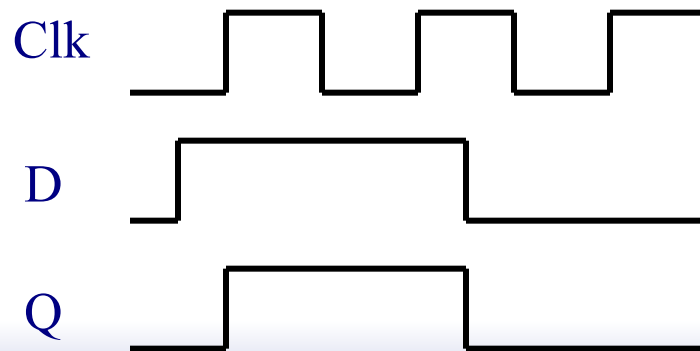
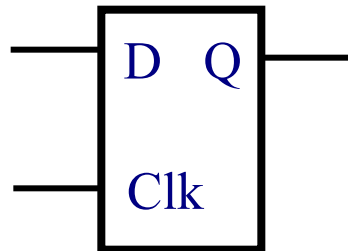
Naming Conventions

- ❑ In our text:
 - a latch is level sensitive
 - a register is edge-triggered
- ❑ There are many different naming conventions
 - For instance, many books call edge-triggered elements flip-flops
 - This leads to confusion however

Latch versus Register

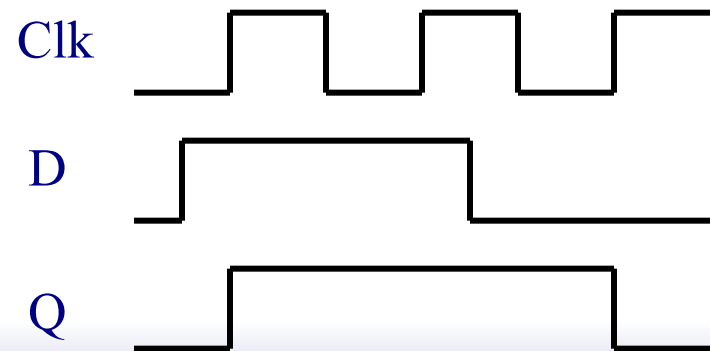
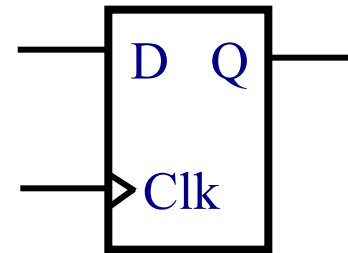
❑ Latch

stores data when
clock is low



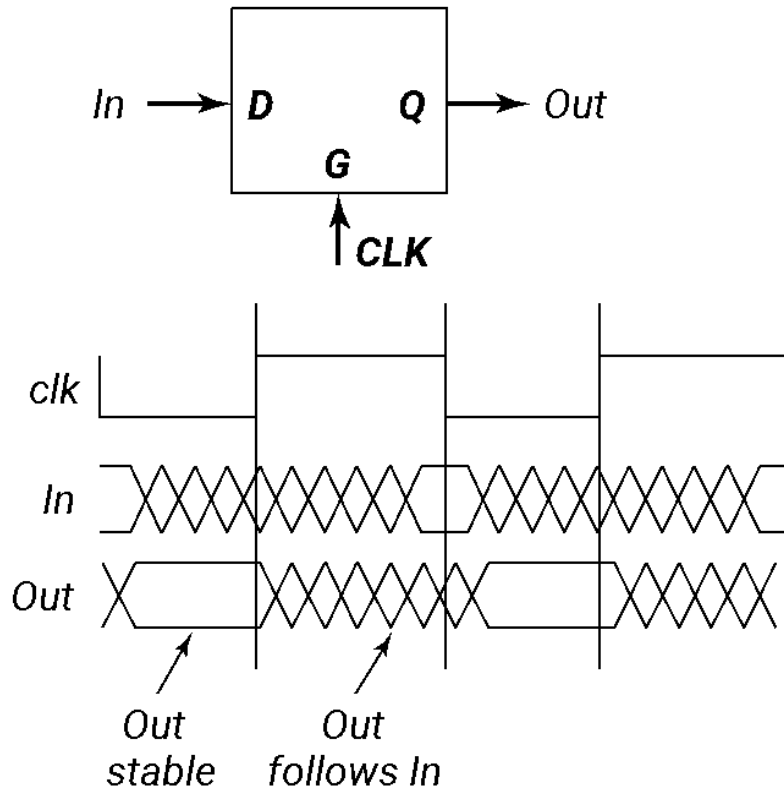
❑ Register

stores data when
clock rises

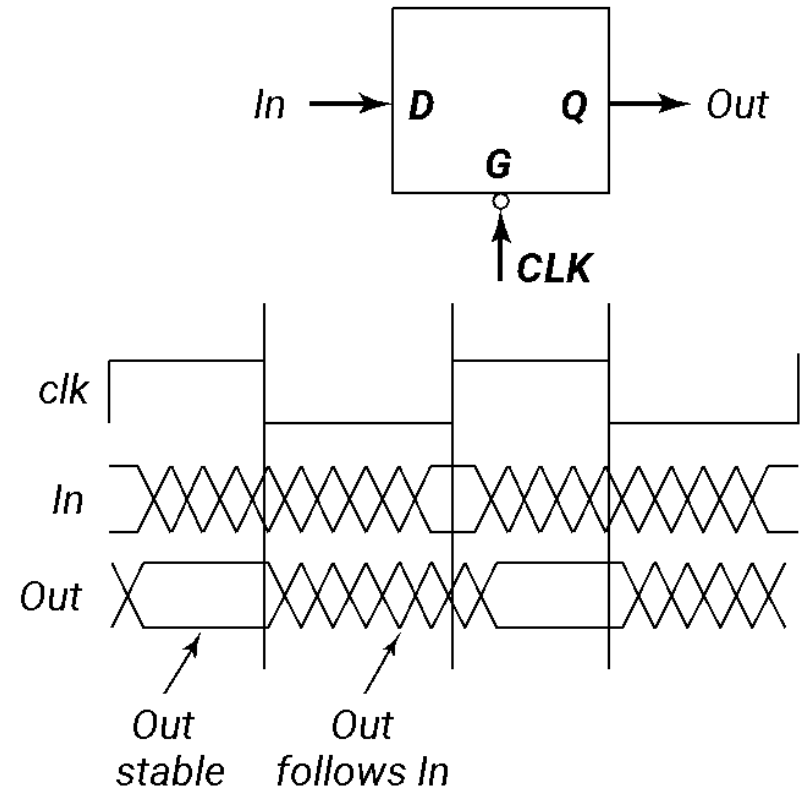


Latches

Positive Latch



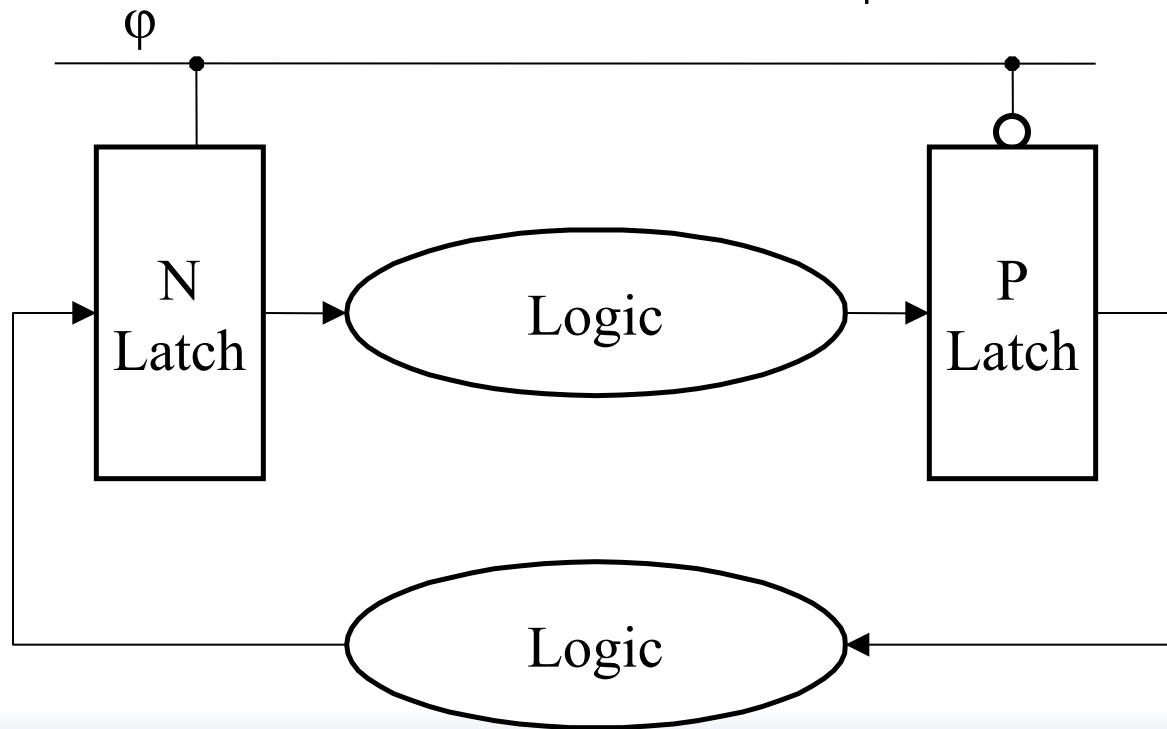
Negative Latch



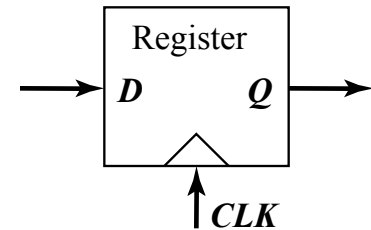
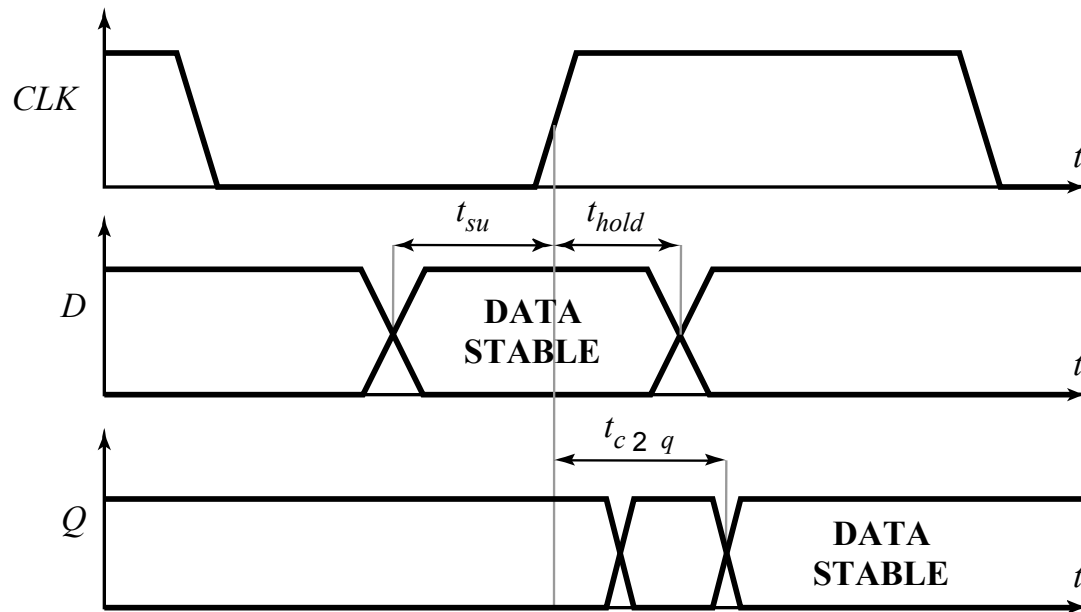
Latch-Based Design

- N latch is transparent when $\phi = 0$

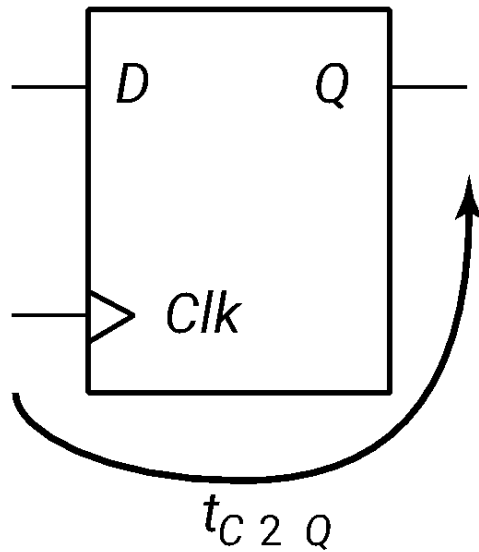
- P latch is transparent when $\phi = 1$



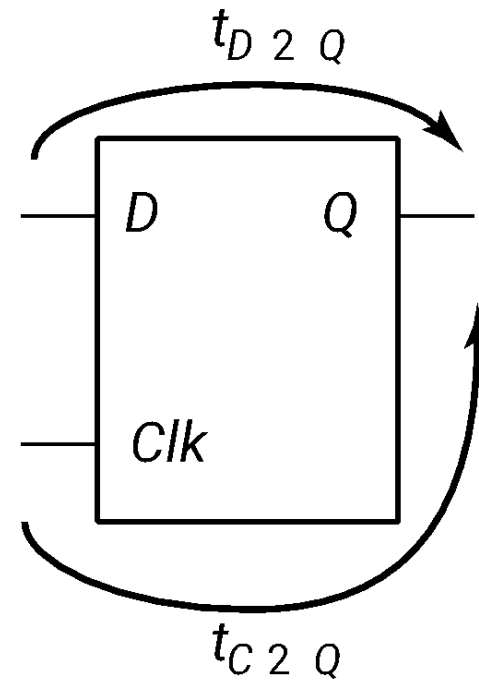
Timing Definitions



Characterizing Timing

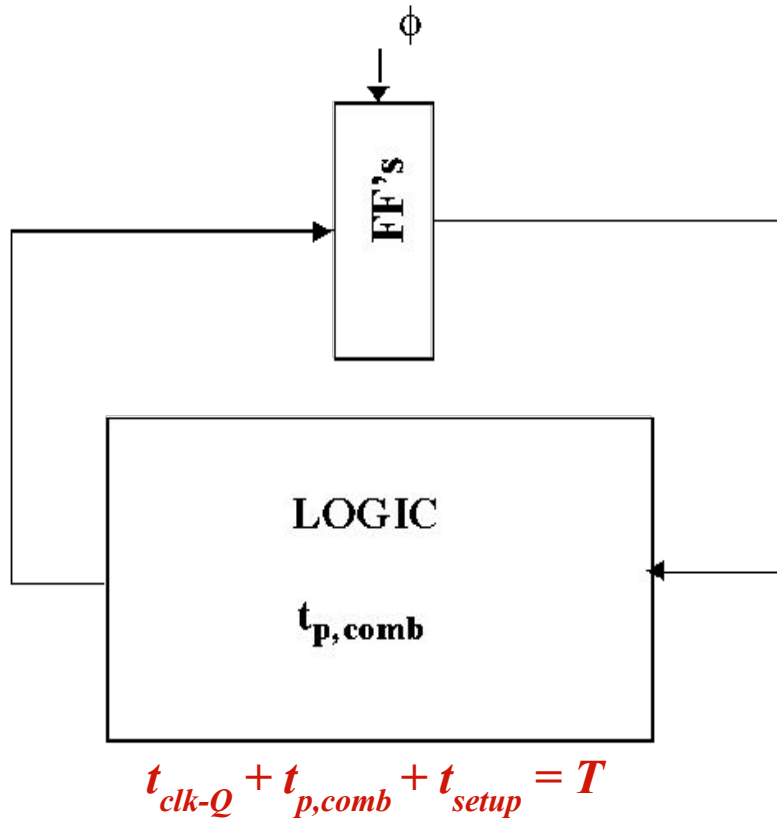


Register



Latch

Maximum Clock Frequency

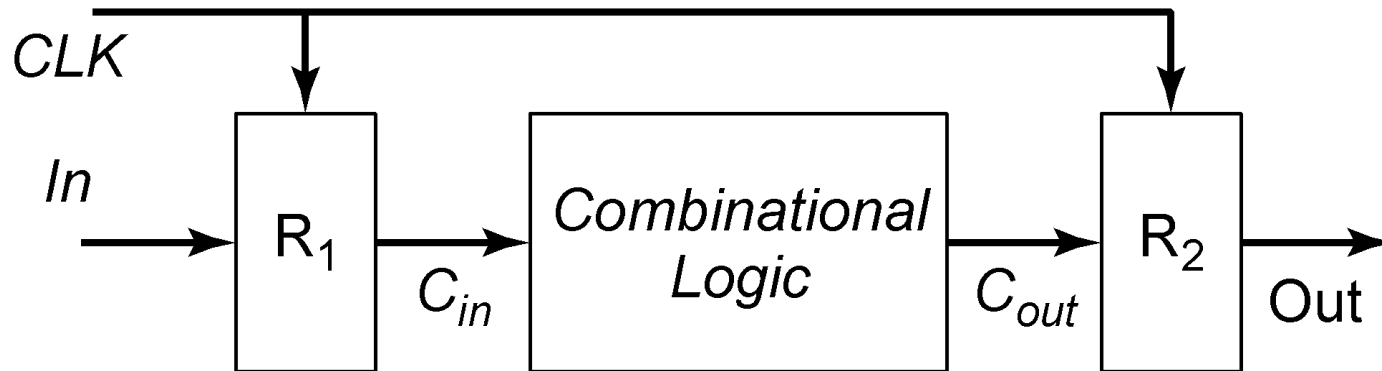


Also:

$$t_{cdreg} + t_{cdlogic} > t_{hold}$$

t_{cd} : contamination delay =
minimum delay

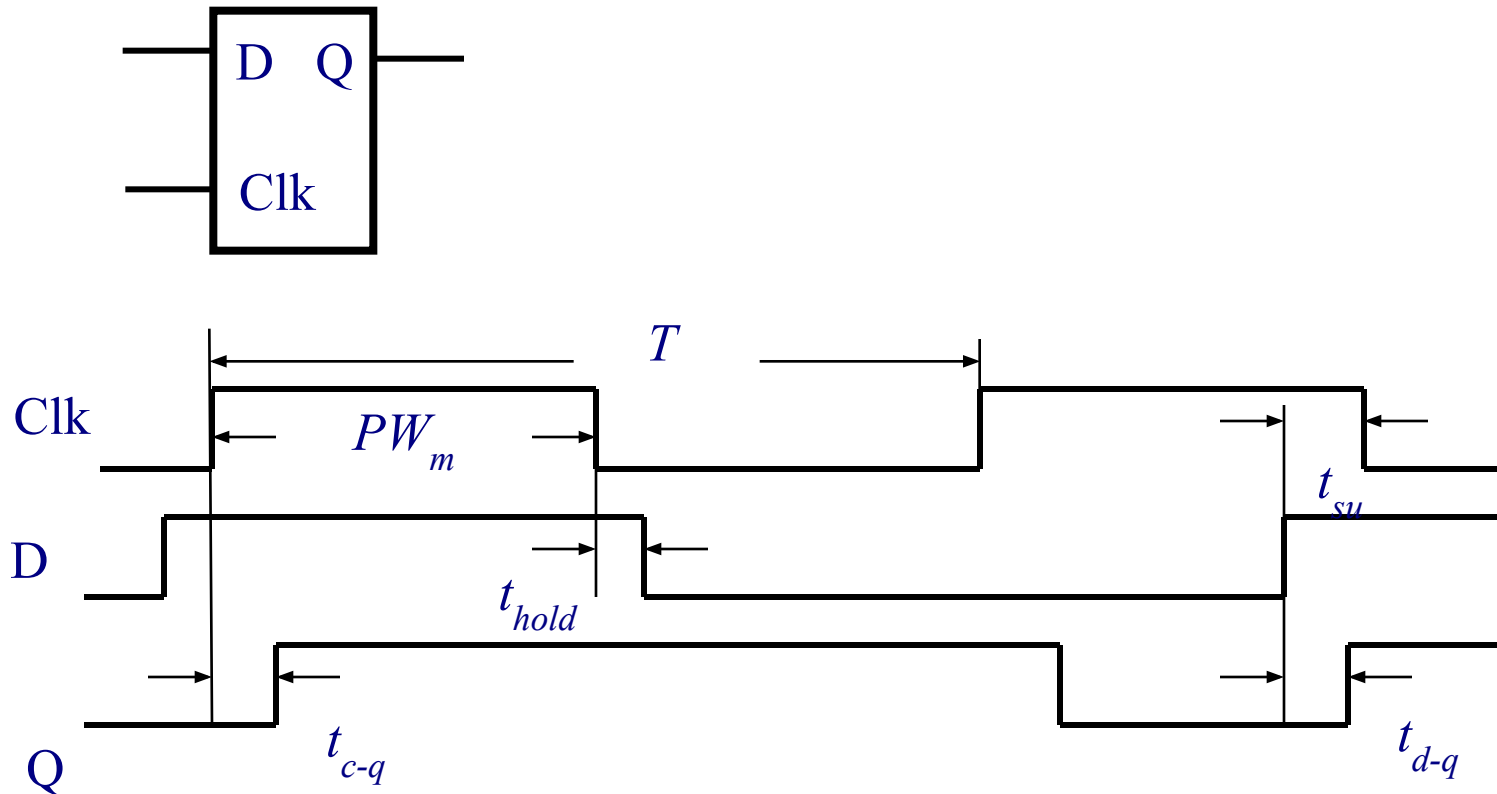
Synchronous Timing





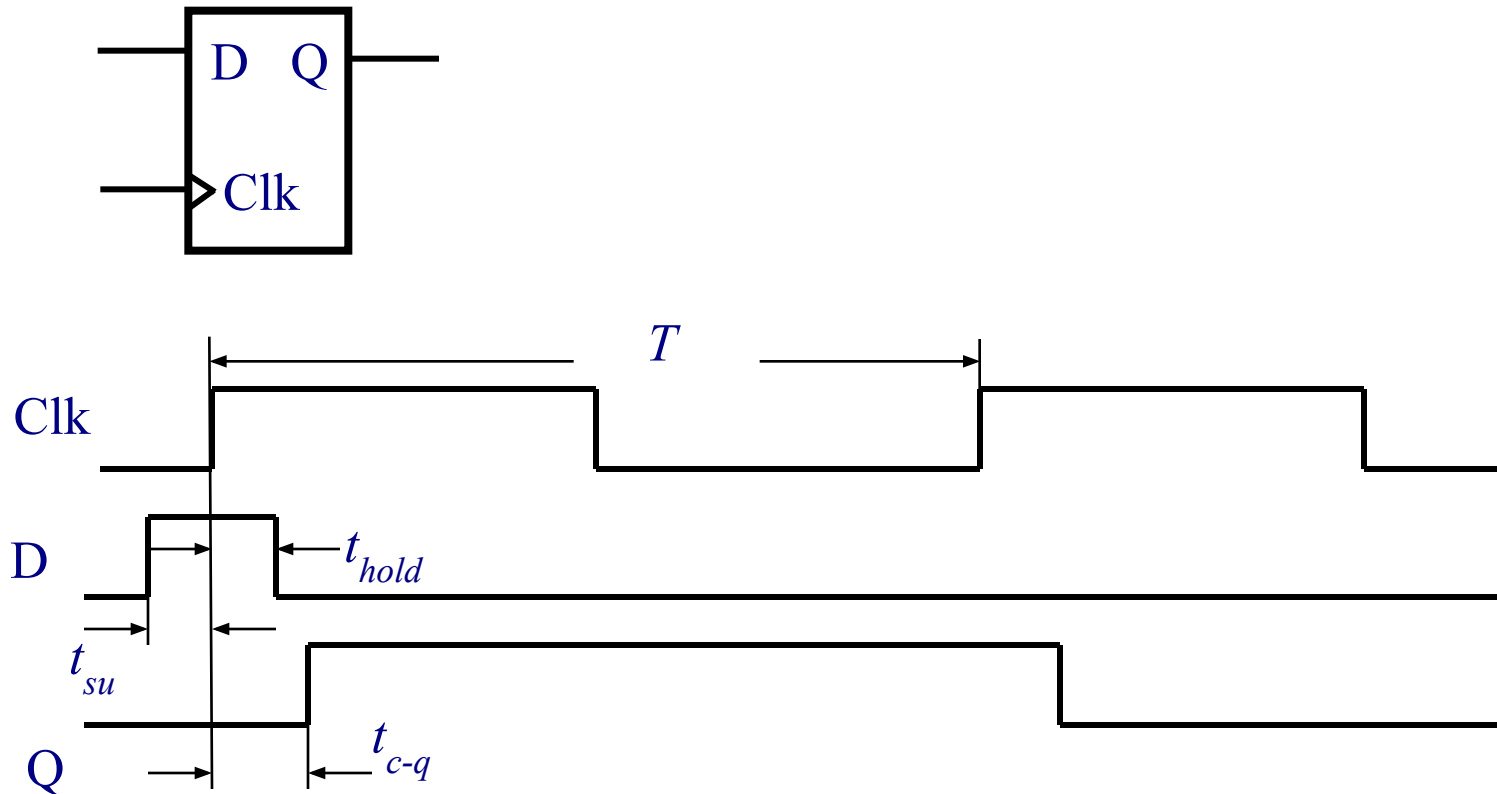
Timing Definitions

Latch Parameters



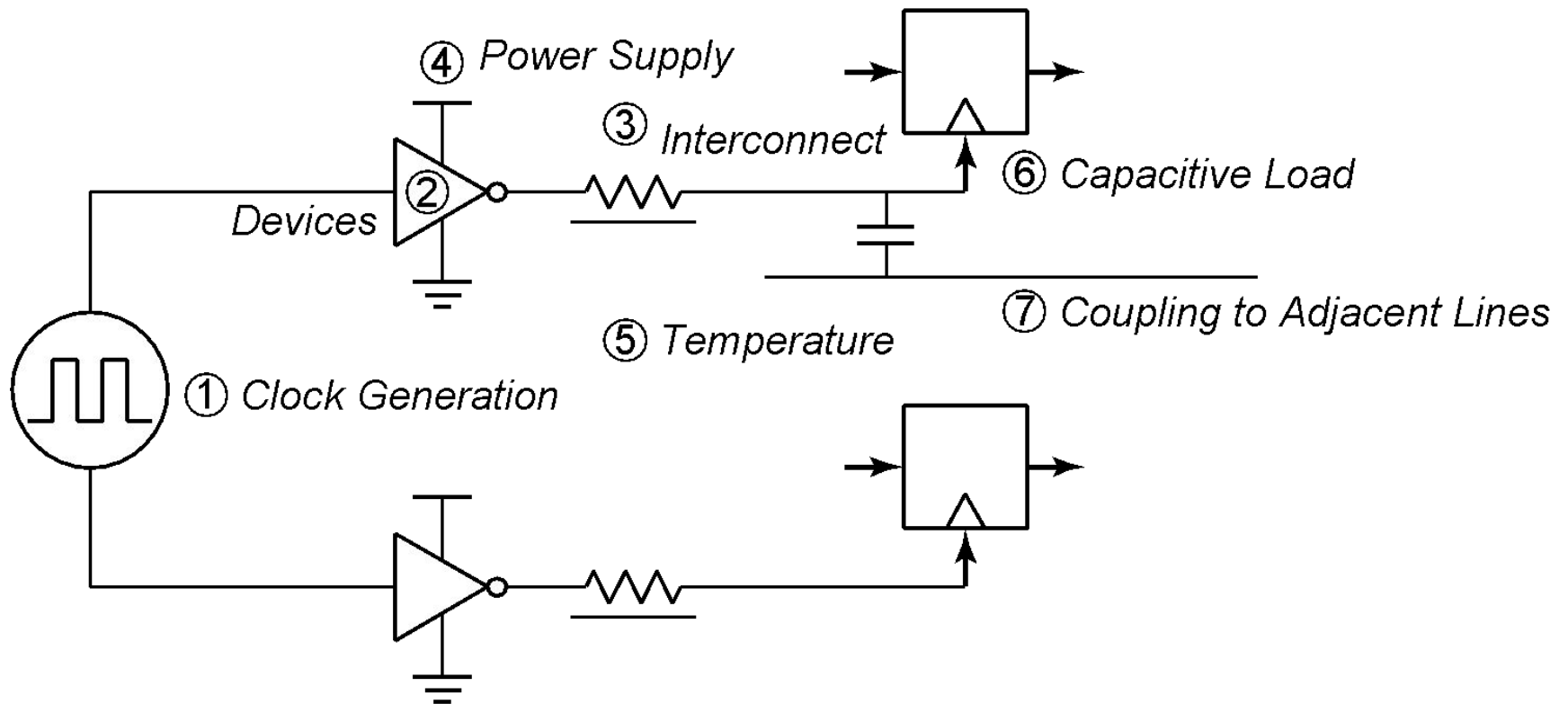
Delays can be different for rising and falling data transitions

Register Parameters



Delays can be different for rising and falling data transitions

Clock Uncertainties



Sources of clock uncertainty

Clock Nonidealities

□ Clock skew

- Spatial variation in temporally equivalent clock edges; deterministic + random, t_{SK}

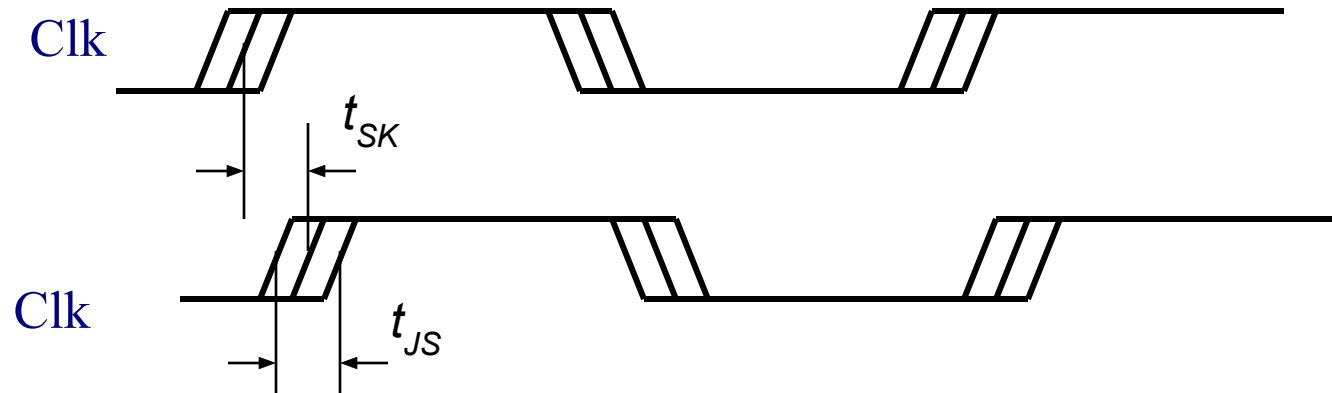
□ Clock jitter

- Temporal variations in consecutive edges of the clock signal; modulation + random noise
- Cycle-to-cycle (short-term) t_{JS}
- Long term t_{JL}

□ Variation of the pulse width

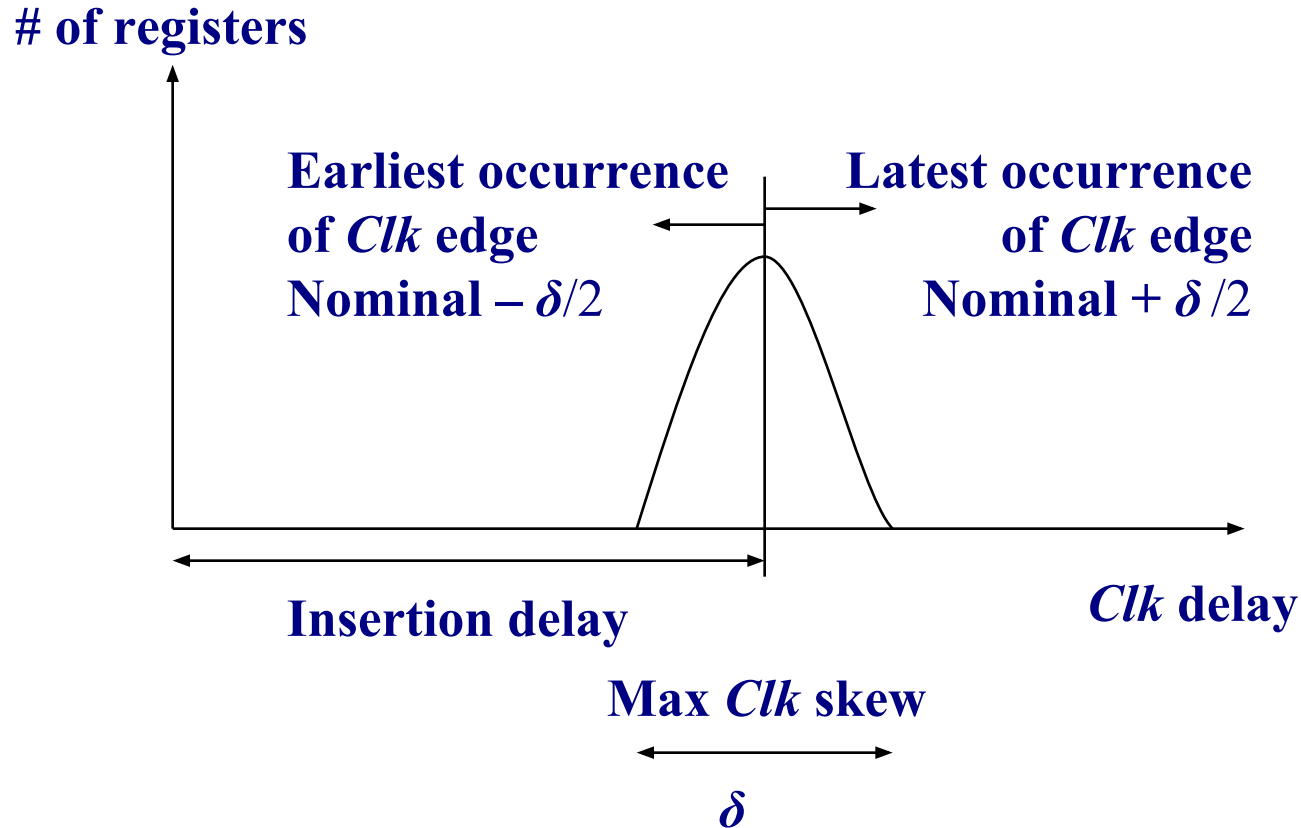
- Important for level sensitive clocking

Clock Skew and Jitter

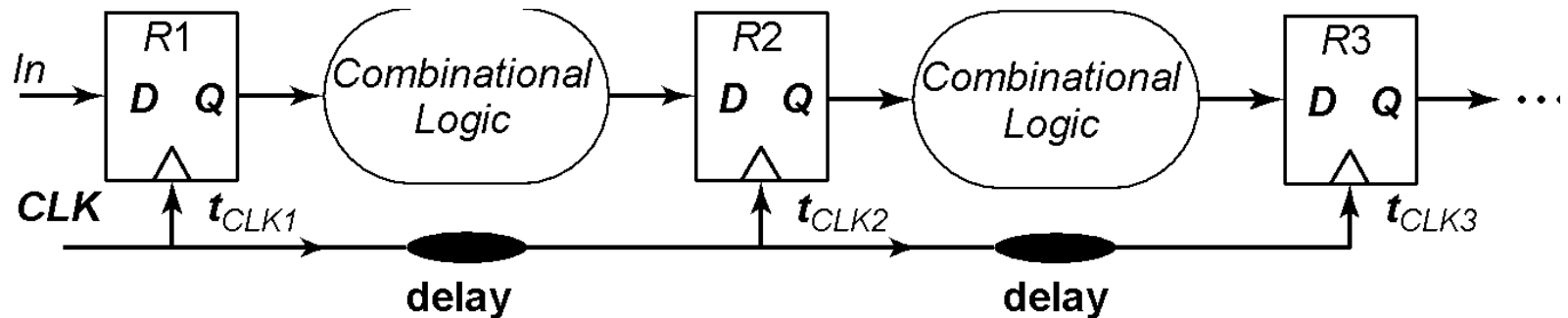


- ❑ Both skew and jitter affect the effective cycle time
- ❑ Only skew affects the race margin

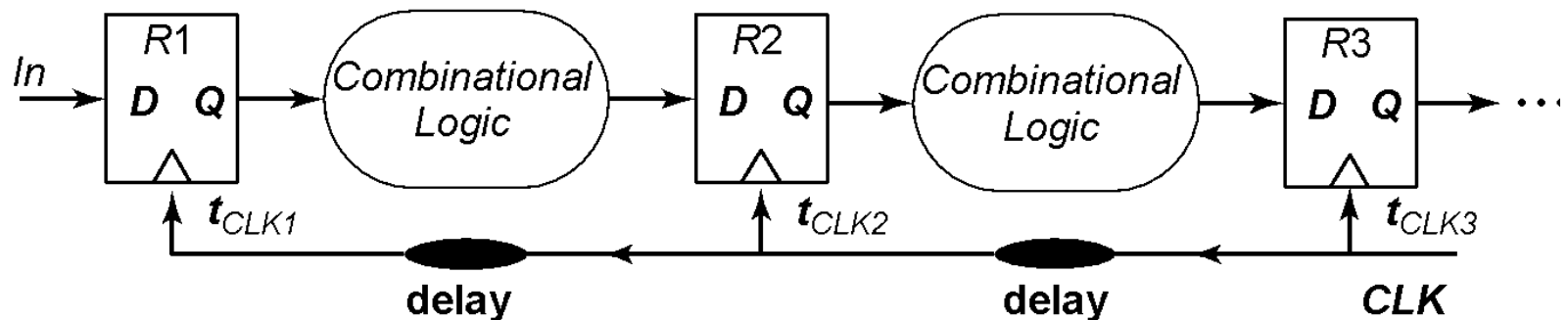
Clock Skew



Positive and Negative Skew

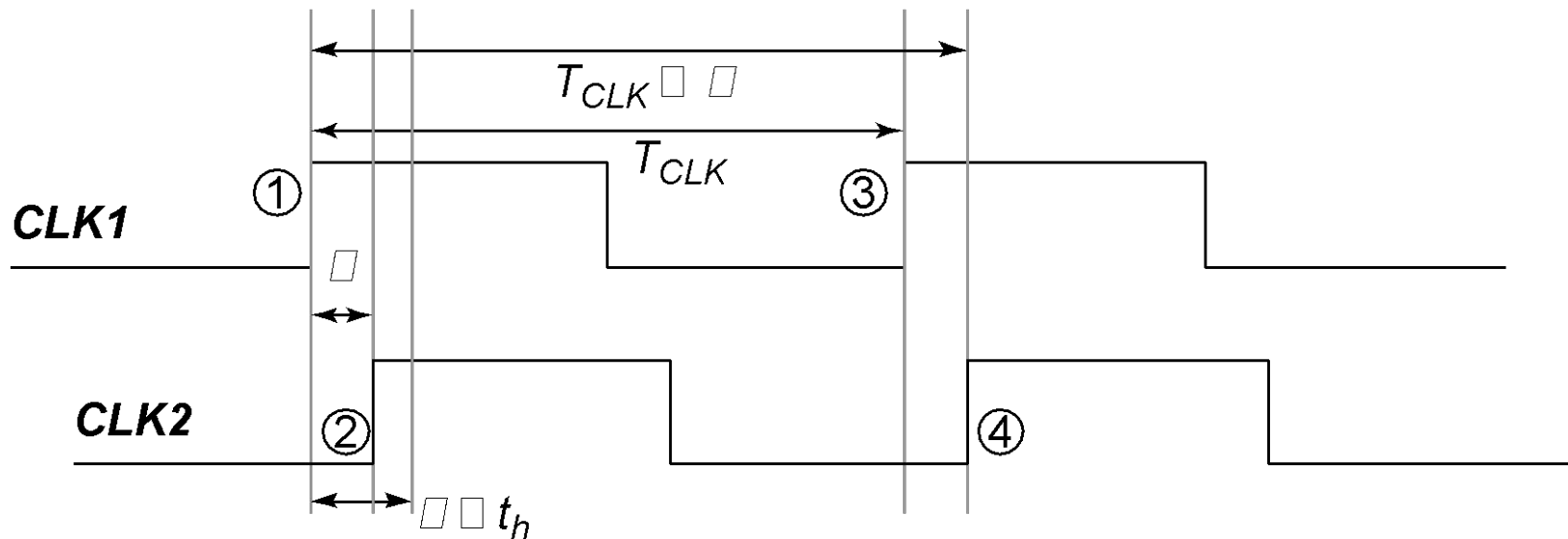


(a) Positive skew



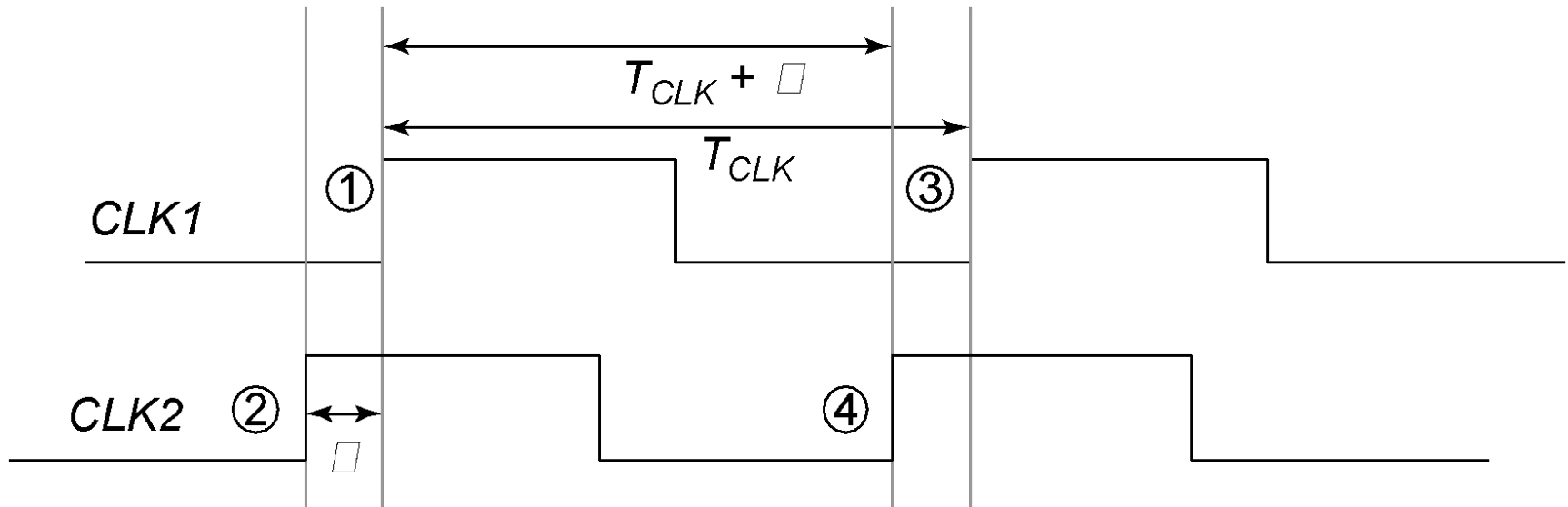
(b) Negative skew

Positive Skew



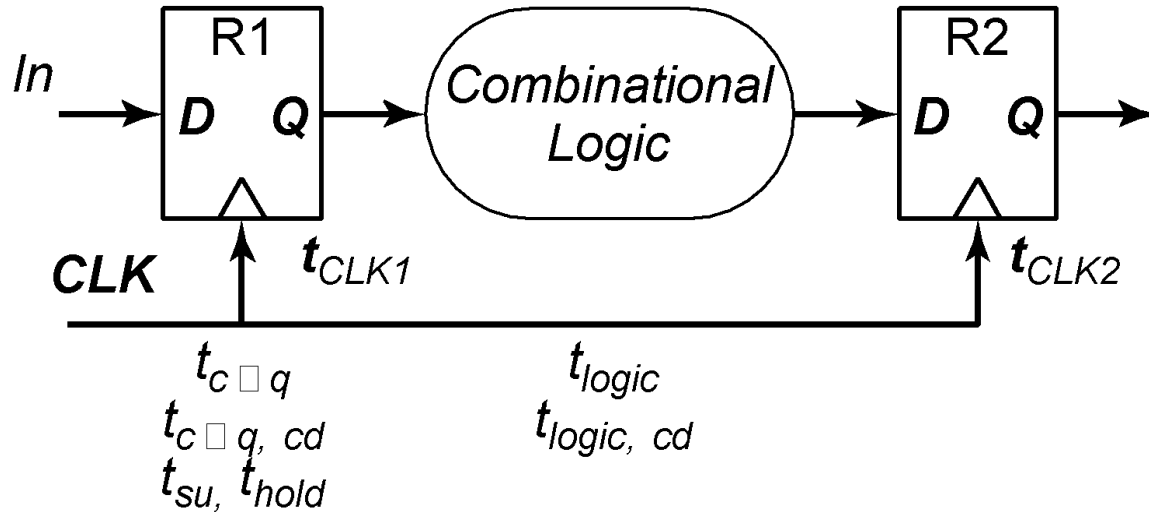
Launching edge arrives before the receiving edge

Negative Skew



Receiving edge arrives before the launching edge

Timing Constraints



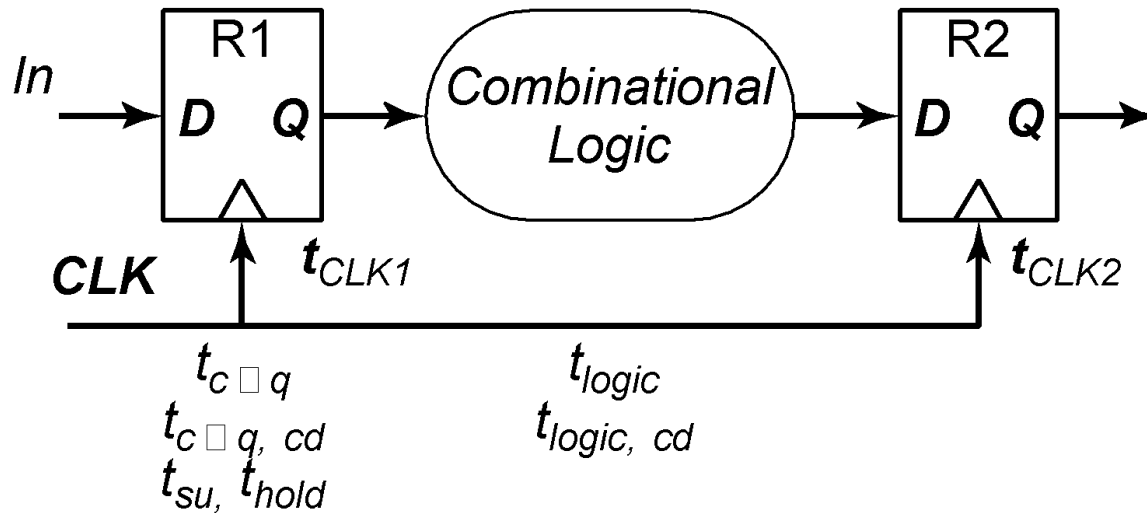
Minimum cycle time:

$$T - \delta = t_{c \rightarrow q} + t_{su} +$$

t_{logic}

Worst case is when receiving edge arrives early (positive δ)

Timing Constraints

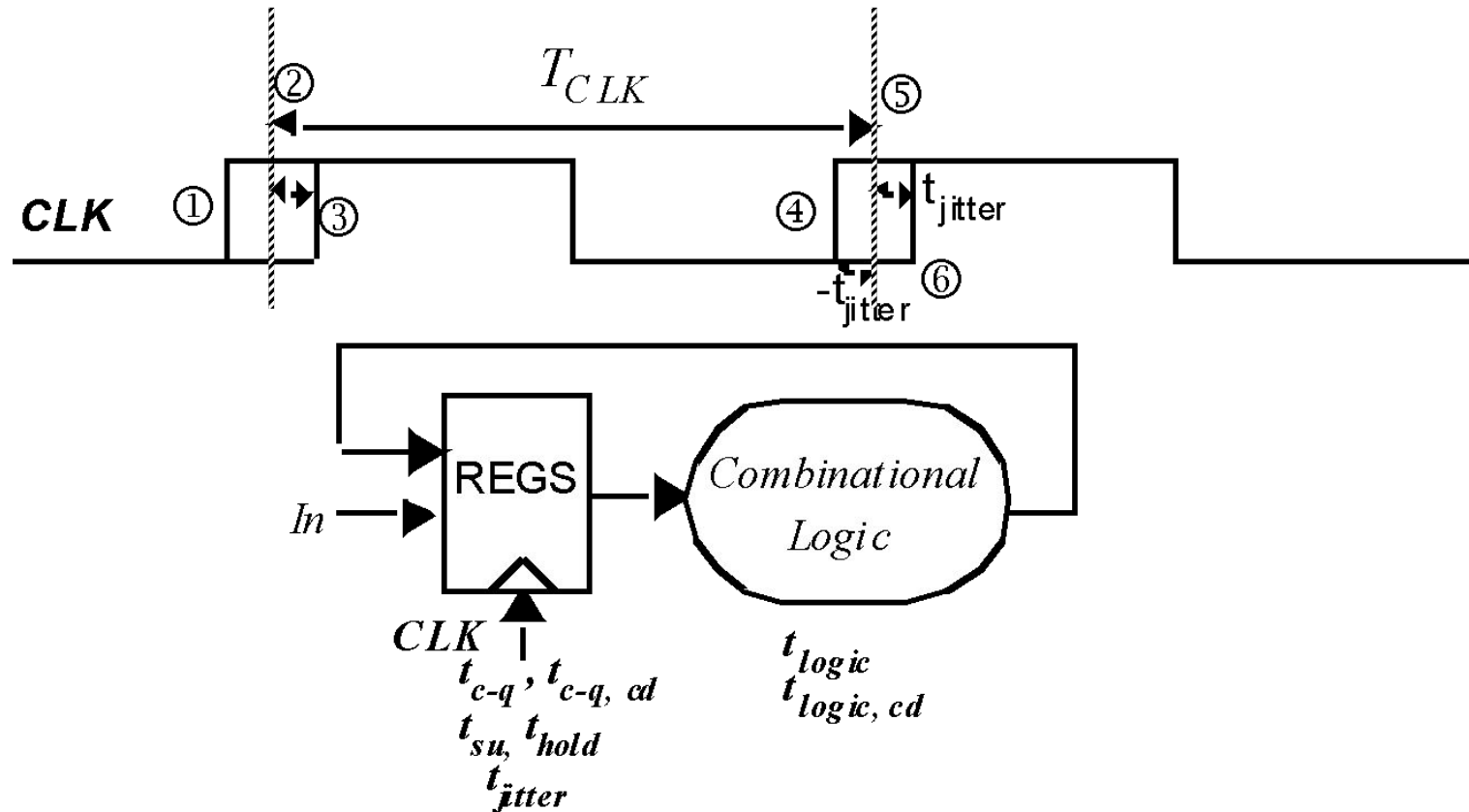


Hold time constraint:

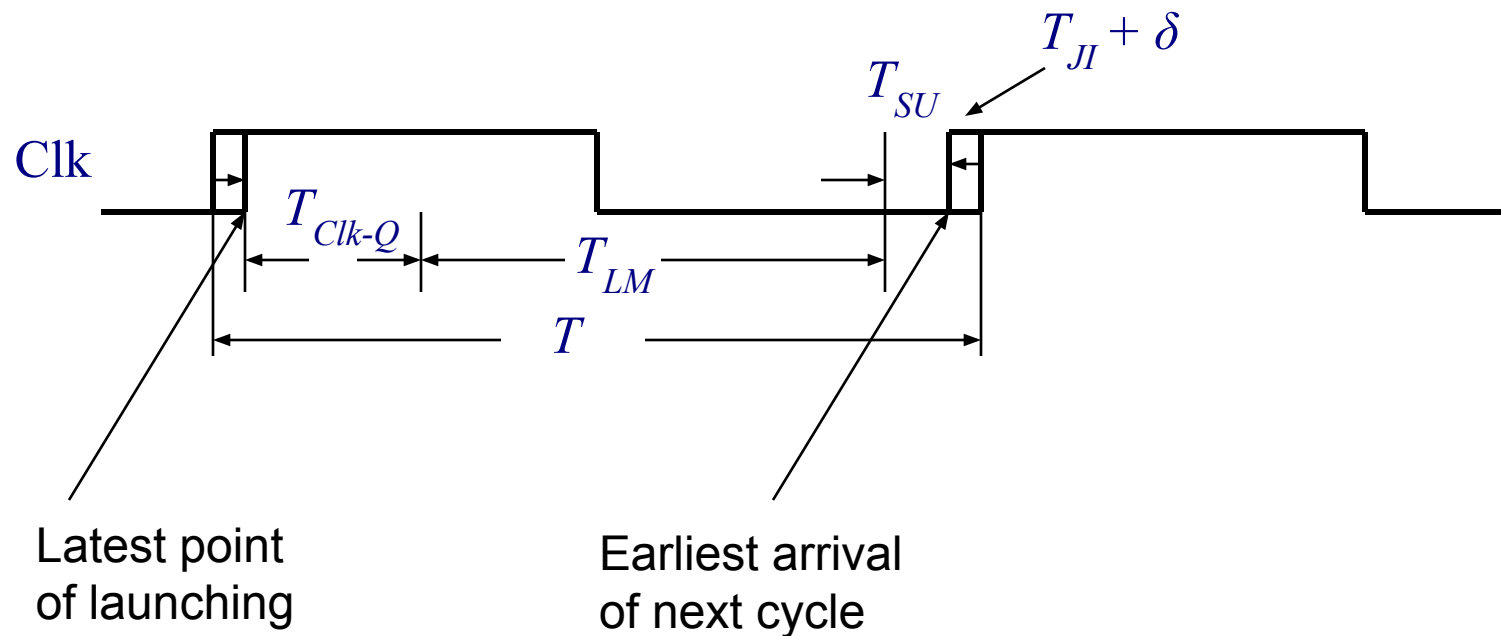
$$t_{(c-q, cd)} + t_{(logic, cd)} > t_{hold} + \delta$$

*Worst case is when receiving edge arrives late
Race between data and clock*

Impact of Jitter



Longest Logic Path in Edge-Triggered Systems



Clock Constraints in Edge-Triggered Systems

If launching edge is late and receiving edge is early, the data will not be too late if:

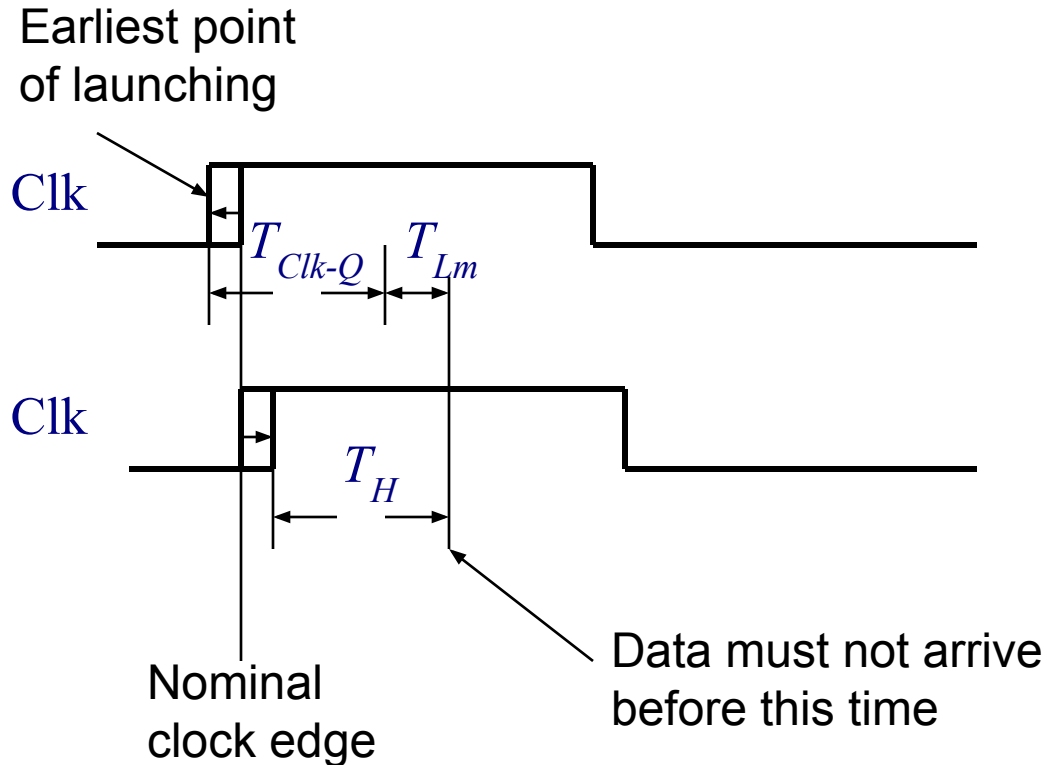
$$T_{c-q} + T_{LM} + T_{SU} < T - T_{JI,1} - T_{JI,2} - \delta$$

Minimum cycle time is determined by the maximum delays through the logic

$$T_{c-q} + T_{LM} + T_{SU} + \delta + 2 T_{JI} < T$$

Skew can be either positive or negative

Shortest Path



Clock Constraints in Edge-Triggered Systems

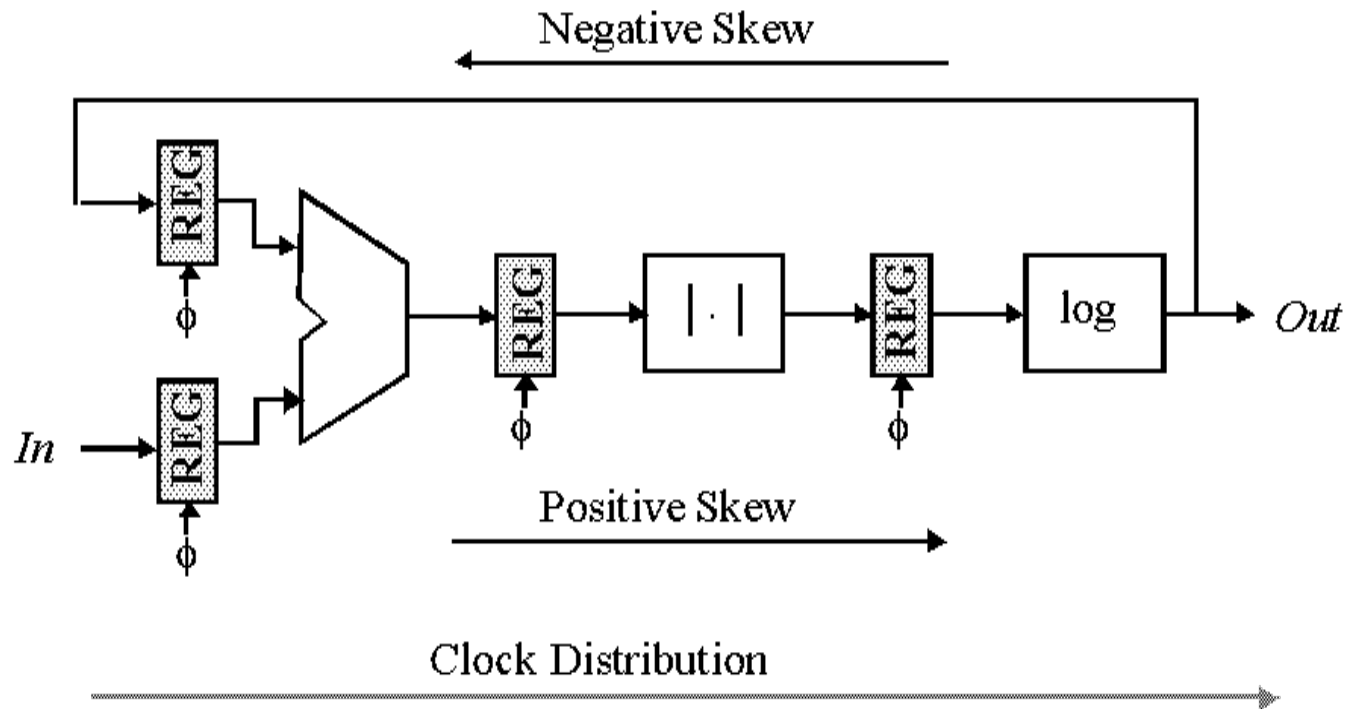
If launching edge is early and receiving edge is late:

$$T_{c-q} + T_{LM} - T_{JI,1} < T_H + T_{JI,2} + \delta$$

Minimum logic delay

$$T_{c-q} + T_{LM} < T_H + 2T_{JI} + \delta$$

How to counter Clock Skew?



Data and Clock Routing